Fairness-Aware Routing in Quantum Networks & Clinical Settings

via Multi-Armed, Contextual, and Informative Bandit Methods

(Rough Draft Report)

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***Abstract*—Sequential decision systems increasingly determine how scarce diagnostic, computational, and communication resources are routed under uncertainty. This project studies fairness-aware contextual multi-armed bandits for such decisions when context is missing, delayed, noisy, or group-dependent. The central claim is that clinical diagnostic decision workflows and quantum network routing share the same structural failure mode: policies must choose an action from partial feedback, yet the groups or flows with the least reliable context can absorb the worst errors. The work develops a reusable evaluation framework for non-contextual bandits, contextual bandits, and informative contextual bandits across two testbeds: a quantum-routing environment grounded in prior hybrid-bandit experiments and a planned simulation-first diagnostic workflow. Work completed so far includes the related-work synthesis, domain model, architecture specification, quantum-first execution plan, and a reproducible CRISP+ML code-review package with unit tests and merge validation. The expected contribution is a reproducible method for measuring and mitigating time-evolving fairness disparities while preserving utility under non-stationarity and limited context.**

I. Introduction

Many applied data science systems are not one-shot predictors. They are sequential decision systems that route limited resources under uncertainty: which diagnostic test to order, which case to escalate, which model or pipeline to trust, or which quantum-network path should receive scarce entanglement-generation attempts. These decisions are difficult because the system observes only partial feedback. The chosen action reveals an outcome, but the unchosen alternatives remain counterfactual. They are also fairness critical because the quality of the observed context is often uneven across groups. When one population has delayed testing, incomplete patient history, lower-quality measurements, or less representative calibration data, a policy optimized only for aggregate reward can hide subgroup errors and compound disparities over time.

This project frames that problem as fairness-aware contextual bandit learning. At each round, a policy observes context, chooses an action, receives reward only for the chosen action, and updates its future decisions. The scientific question is not only whether contextual information improves reward, but whether informative context and fairness-aware monitoring reduce disparities when context is missing or degraded. This matters in clinical workflows because false-negative or delayed-escalation gaps can produce direct harm. It also matters in quantum-network routing because future communication systems will allocate scarce probabilistic resources among competing flows; fairness in success probability and latency is a service-quality issue rather than a purely normative add-on.

The project uses two complementary testbeds. The first is an existing quantum-routing evaluation stack based on hybrid contextual bandits for entanglement routing and qubit allocation [1]. That testbed already exposes uncertainty, adversarial or non-stationary conditions, and stateful evaluator artifacts. The second is a planned diagnostic-like simulation motivated by equitable bioinformatics and fairness auditing work [2]. The diagnostic simulation will model test selection, retesting, and escalation under controllable missingness, measurement noise, and group-dependent context quality. The two-domain design tests whether fairness mechanisms and policy-update strategies transfer across fundamentally different applications.

The current rough draft reports the mostly completed design and implementation foundation rather than final experimental results. The remainder of the paper is organized as follows. Section II summarizes related work. Section III describes the methodology and domain model. Section IV describes the implementation and reproducibility architecture. Section V summarizes the data sets and experimental inputs. Section VI gives preliminary validation evidence. Sections VII and VIII describe future work and conclusions.

The rough-draft contribution is therefore not a claim that the final fairness results are complete. Instead, it consolidates the work into a single preliminary final-paper structure. That consolidation is important because the project is cross-domain. The organizing idea used throughout this draft is that both domains are sequential routing systems whose fairness risk emerges from unequal context quality. The clinical system routes diagnostic attention; the quantum system routes probabilistic network resources. Both require learning under partial feedback, non-stationarity, and subgroup- or flow-level performance monitoring.

II. Related Work

A. Bandits and Contextual Sequential Decision-Making

Multi-armed bandits model online learning under partial feedback, balancing exploration and exploitation while observing rewards only for selected actions. UCB-style algorithms provide finite-time regret guarantees in stochastic settings [3], while Thompson sampling offers a Bayesian alternative that often performs well empirically [4]. The broader bandit literature provides the regret and uncertainty-estimation foundation for this project [5]. Contextual bandits extend this model by conditioning action choice on observed side information. LinUCB and related methods demonstrate how context can improve online personalization and decision quality [6], [7], and reduction-based contextual bandit approaches show how richer policy classes can be scaled [8]. However, most of this work optimizes expected reward and assumes that context is observed correctly or representatively.

Offline policy evaluation is also relevant because the final framework will compare policies from logged simulator outputs as well as from fresh online runs. Doubly robust estimators combine reward modeling with inverse-propensity weighting and can reduce variance when evaluating bandit policies from logged data [9]. In this project, those ideas motivate careful logging of actions, propensities or selection rules, context snapshots, and oracle comparisons. They also highlight a fairness concern: if one group has less support in the logged data, group-specific estimates can be noisier even when the aggregate estimator appears stable.

B. Fairness, Missing Context, and Non-Stationarity

Fairness in supervised learning is often expressed through group-conditioned error criteria such as equality of opportunity and equalized odds [10]. In sequential settings, the problem is harder: fairness constraints affect exploration, data collection, and the future evidence available to the learner. Joseph et al. connect fairness to bandit learning directly [11], while Chen et al. study fair contextual multi-armed bandits with explicit group constraints [12]. Offline contextual-bandit fairness work shows how importance-weighted estimators can support high-probability constraints [13], but online distribution shift and time-evolving disparity remain open practical challenges. Non-stationary bandit work addresses changing reward distributions through discounting, change detection, or variation budgets [14], [15]. Limited or unobserved context work similarly motivates algorithms that remain robust when features are unavailable or latent [16], [17].

The limitation across these areas is that they often address only one axis of the final problem. Standard bandits provide regret machinery but not fairness audits. Fairness definitions clarify what disparities matter but are often studied in static supervised settings. Non-stationary bandits handle changing reward processes but usually report aggregate utility. Restricted-context work handles missing features but rarely treats missingness as a group-dependent fairness mechanism. The present project combines these axes by treating missing context, time-varying reward, and group disparity as simultaneous properties of the same learning process.

TABLE I
How the reviewed literature supports the project.

Literature area	Role in this rough draft
Bandit foundations	define partial feedback, exploration, regret, and baselines
Contextual bandits	model side information and missing-context sensitivity
Fairness definitions	specify outcome gaps such as FNR/FPR and service equity
Non-stationarity	motivates shift scenarios and time-evolving disparity curves
Clinical bias studies	motivate subgroup error monitoring and proxy-risk caution
Quantum routing	provides resource-scarce routing testbed and stateful simulator

C. Clinical Fairness and Quantum Routing

Clinical algorithmic bias studies show that proxy targets and measurement processes can encode structural inequities, producing under-allocation for some groups even when aggregate performance appears strong [18]. This motivates outcome-based disparity monitoring such as false-negative and false-positive gaps. In parallel, quantum-network research describes an emerging communication infrastructure in which routing decisions are governed by probabilistic entanglement generation, fidelity, storage limits, and scarce resources [19]–[21]. Prior quantum routing work usually emphasizes throughput, fidelity, or robustness rather than group-based service equity. The gap this project addresses is therefore the intersection of contextual bandit learning, missing context, non-stationarity, and explicit fairness monitoring across both clinical and quantum-resource routing domains.

Table I summarizes the role of each category. The final paper will expand this discussion with direct comparisons once the final experiments are complete, but the rough draft already establishes the scientific gap: existing work rarely evaluates outcome-based fairness under missing context and distribution shift across two structurally similar but domain-distinct routing systems.

III. Approach and Methodology

A. Shared Missing-Context Routing Model

The methodology treats both domains as routing under missing context. In the clinical domain, the routed resource is diagnostic attention: tests, retests, triage escalation, or model/pipeline selection. In the quantum domain, the routed resource is entanglement-generation effort or path capacity. In both domains, the policy observes partial context, chooses an action, receives partial feedback, and updates. The key experimental variable is the richness and reliability of context available to the policy.

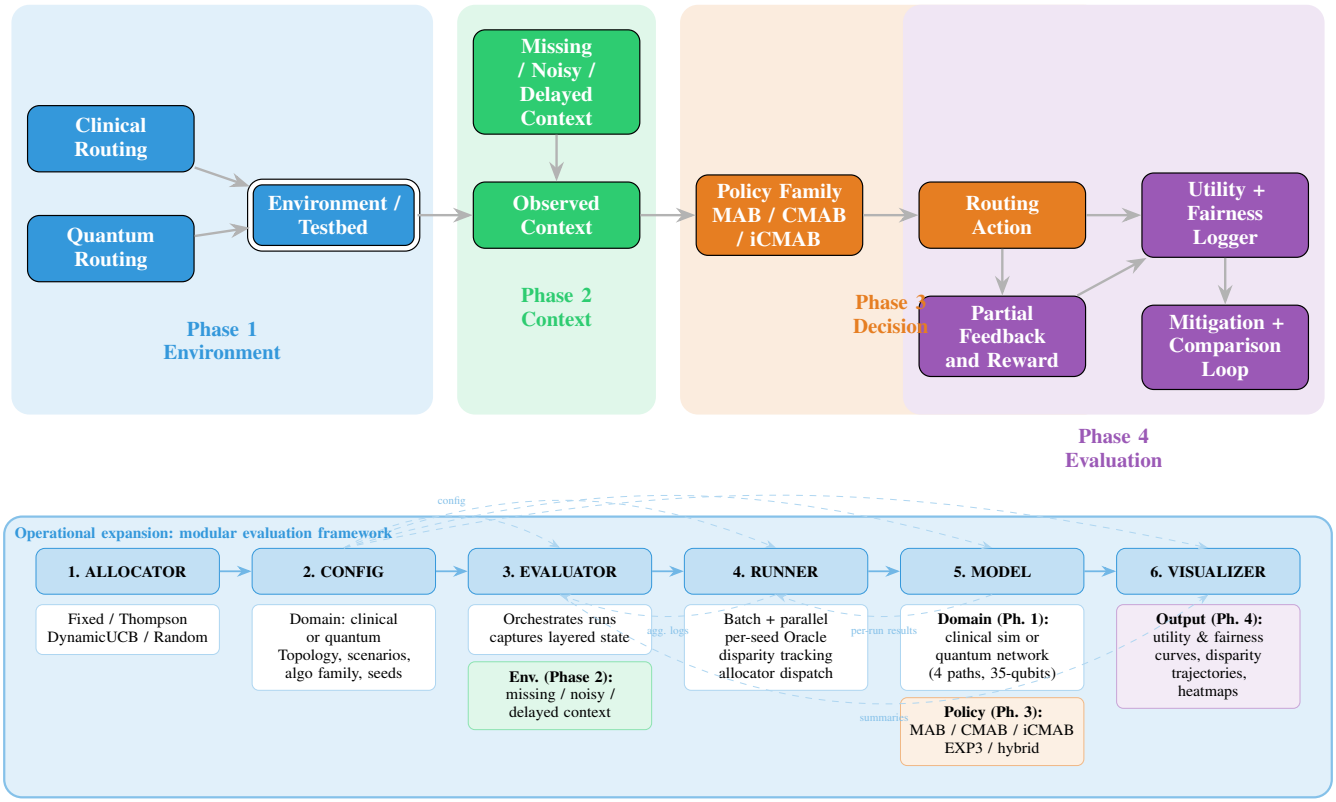

Fig. 1. Shared domain model and operational evaluation framework adapted from the approach write-up. The top panel shows the four project phases: environment/testbed, context observation, routing decision, and evaluation/mitigation. The bottom panel expands the testbed interface into the allocator, configuration, evaluator, runner, model, and visualizer layers used to run both clinical simulation and quantum routing experiments.

Figure 1 summarizes the shared model. Phase 1 defines the environment and available resources. Phase 2 determines what context is observed and which context is missing or degraded. Phase 3 selects an action using a non-contextual, contextual, or informative contextual bandit policy. Phase 4 evaluates both performance and fairness, then feeds mitigation or updated policy information back into future decisions. This model is useful because it makes the failure point explicit: if context is systematically lower-quality for one group or flow class, the learning process can convert measurement inequity into repeated allocation inequity.

At round t, the learner receives an observed context vector $\tilde{\mathbf{x}}_t$, which may differ from the latent complete context $\mathbf{x}_t$ because of missingness, delay, or measurement noise. It chooses an action a_t and observes reward $r_t(a_t)$ only for that action. Utility is tracked through cumulative regret,

$$R_T = \sum_{t=1}^{T} \left(r_t(a_t^*) - r_t(a_t) \right), \tag{1}$$

where a_t^* is the best available action under the simulator oracle. Fairness is tracked through time-indexed disparity gaps,

$$\Delta_m(t) = \left| m_t(G=0) - m_t(G=1) \right|, \tag{2}$$

where m_t is a domain-specific outcome metric such as false-negative rate, success probability, latency, or escalation delay. This notation separates the learning objective from the fairness audit: a policy can have low regret in aggregate while still producing high $\Delta_m(t)$ for specific groups.

B. Design Decisions

Bandits are used instead of full reinforcement learning because both testbeds provide bandit feedback: only the chosen diagnostic or routing action reveals an outcome. The project compares three policy families. Non-contextual MABs serve as a baseline for action-only learning. Contextual bandits test whether observed context improves utility and equity. Informative contextual bandits represent policies that recover, augment, or better weight missing context. The simulation-first clinical design avoids sensitive patient data while allowing controlled missingness and group-specific measurement noise. The quantum testbed reuses an existing, domain-grounded routing environment so that fairness analysis is built on top of realistic uncertainty rather than a toy network.

Fairness is measured as an online outcome property. In diagnostics, the target metrics are group-wise false-negative and false-positive gaps, along with utility and cost-sensitive reward. In quantum routing, service equity is measured through success-probability and latency gaps across flow groups. Mitigation mechanisms under consideration include fairness-regularized policy updates, group-aware calibration or thresholding, and missingness-aware context augmentation. These mechanisms

will be evaluated as tradeoffs rather than as free improvements, because fairness constraints can reduce short-term reward while improving reliability and equity.

C. Experimental Questions

The final experiments will answer four questions. First, how much does context improve utility and fairness relative to non-contextual baselines when all groups have comparable context quality? Second, how quickly do fairness gaps emerge when one group or flow class receives noisier, delayed, or incomplete context? Third, do missingness-aware or fairness-regularized policy updates reduce outcome disparities without unacceptable utility loss? Fourth, do the same policy-update strategies transfer between clinical diagnostic simulation and quantum routing, or are the fairness mechanisms domain-specific? These questions define the planned analysis tables and figures: reward/regret curves, time-evolving disparity curves, mitigation tradeoff plots, and cross-domain transfer summaries.

D. Planned Mitigation Strategy

The first mitigation family will treat fairness as a penalty or constraint in the policy-update loop. In the simplest form, the learner receives a utility reward and a disparity signal; policy updates are then adjusted when recent group-specific outcomes exceed an allowed gap. This makes fairness operational during learning instead of applying it only after the experiment. A second family will use missingness-aware context augmentation. If a group or flow class systematically receives incomplete context, the policy can either acquire additional features, impute missing indicators, or adjust confidence estimates so that uncertainty caused by measurement gaps is not mistaken for low expected value. A third family will test group-aware calibration or thresholding at the decision boundary. This is especially relevant for diagnostic-like escalation, where a small threshold shift can change false-negative rates.

The mitigation comparison will be conservative. The final paper will not claim that lower disparity is automatically better if the policy produces unsafe or unusable utility loss. Instead, each mitigation will be evaluated through utility–fairness tradeoff curves. The preferred policy is expected to be one that preserves most of the utility of contextual learning while reducing time-evolving disparity under missing context. This is why the project compares MAB, CMAB, and iCMAB policies under the same seeds, scenario settings, and logged artifacts: the point is to isolate the effect of context quality and fairness-aware updates rather than confounding algorithm choice with simulator configuration.

IV. Implementation and Reproducibility

A. Execution Paths

The implementation is designed around reproducibility. The quantum side uses three execution paths inherited from the upstream quantum project: notebooks for fast interactive replication, a local environment for development and testing, and GCP/batch scripts for longer runs. The notebook path supports Paper2, Paper7, Paper8, and Paper12-style testbed configurations. The local path uses a virtual environment, dependencies from `requirements.txt`, and validation scripts. The GCP path uses startup and dynamic-runner scripts for larger allocator and time-scale experiments.

TABLE II
Implementation layers and reproducibility role.

Layer	Role
Configuration	fixed seeds, scenario settings, run scales, and output roots
Collection / state access	retrieves shared Drive framework and model states
Runner / allocator	executes quantum routing actions and qubit allocation
Evaluator	computes scenario winners, reward, efficiency, and gap summaries
Visualizer / analysis	generates plots, summaries, and review-facing tables
CRISP+ML package	provides small deterministic pipeline, tests, and merge checks

Table II summarizes the software entities. The quantum workflow relies on evaluator and runner objects that save framework state, model state, and analysis outputs. These state layers support interruption recovery and replication. The CRISP+ML package in its own repository is a reviewer-safe implementation harness that mirrors the project methodology: data collection, cleaning, preparation, exploration, mining, evaluation, and postprocessing. It exposes command-line stages through `python -m experiments`, writes deterministic artifacts under a result directory, and records provenance through a pipeline manifest.

The state design is intentionally layered. Individual models can save model state so interrupted algorithm runs do not always require recomputation. Experiment runners can save complete experiment state so a particular scenario, seed, allocator, and capacity setting can be resumed. Evaluators can save scenario-level collections of runner outputs so a full comparison can be extended, summarized, or audited later. This layered persistence is important for research validity because long-running quantum-routing jobs can be interrupted, and because later fairness analysis may need to reconstruct how a specific result was produced. In the final system, allocator-runner state will become a fourth layer that aggregates all evaluator runs for a specific allocator and supports broader cross-scenario analysis.

Table III shows how the small code package maps to a recognized methodology rather than an ad hoc script sequence. The same conceptual pipeline will be used for the final experiments: collect or generate simulator records, clean and validate them, prepare model-ready contexts, run policy families, evaluate utility and fairness, and present results in reproducible plots and tables.

B. Code Review and Validation Path

The code artifact is intentionally object-oriented. A base pipeline-stage interface defines the stage contract, concrete

TABLE III
CRISP+ML MAPPING USED BY THE REVIEWER-SAFE IMPLEMENTATION.

Lifecycle stage	Concrete artifact
Data collection	synthetic fairness-style raw records and metadata
Data cleaning	validated, normalized records with missingness handling
Data preparation	train/test artifacts and model-ready feature table
Data exploration	summary statistics and group/context diagnostics
Data mining	lightweight deterministic model or bandit-style run
Evaluation	utility metrics and group-disparity summaries
Presentation	final table, manifest, and visualization artifact

classes implement each CRISP+ML phase, typed configuration and artifact containers carry paths and metadata, and a pipeline orchestrator runs stages end-to-end or stage-by-stage. Unit tests validate each stage and the overall manifest. The merge-check command runs the full quick-review pipeline and the unit-test suite, giving reviewers a fast way to verify that the repository executes before inspecting the larger quantum workflow. This design supports the rough-draft requirement that another party should be able to recreate the system or software from the description.

The package also supports stage-level commands rather than requiring reviewers to run notebooks. Each stage can be executed independently, and the full pipeline command composes those same objects. This separation makes failures diagnosable: a schema error should fail in cleaning, a feature construction problem should fail in preparation, and a missing metric should fail in evaluation. The GitHub merge-check workflow mirrors the local validation path, so code submitted for review is checked against the same unit tests and quick-run pipeline that a reviewer would execute manually.

C. Recreation Path for Reviewers

A reviewer should be able to recreate the small software artifact without private data or notebooks. The intended path is: create a Python virtual environment, install dependencies from `requirements.txt`, run the stage-level commands if inspecting individual lifecycle steps, and run the local merge-check command before accepting changes. The code-review repository therefore acts as a miniature version of the larger research system. It does not attempt to reproduce every long-running quantum experiment; instead, it proves that the lifecycle, object contracts, validation style, and fairness-oriented metric outputs are executable in a small deterministic setting.

The larger quantum workflow remains available through notebooks, local scripts, and GCP execution. The notebooks are useful for rapid inspection and for matching earlier Paper2, Paper7, Paper8, and Paper12 testbed conventions. The local path is useful for debugging evaluator, runner, visualizer, and state-management behavior. The GCP path is reserved for long jobs that would be too slow for local review. Shared Drive storage is part of the immediate execution model because it allows state files, model outputs, and analysis artifacts to persist across machines and interrupted sessions. This is especially important for quantum runs whose value is not only the final metric but also the sequence of states and decisions that produced it.

V. Data Sets and Experimental Inputs

The current quantum testbed uses simulated quantum-network routing scenarios with probabilistic link success, qubit-capacity constraints, allocator settings, time scales, and stochastic or adversarial conditions. The key outputs are framework-state and model-state artifacts, reward traces, oracle comparisons, winner summaries, efficiency measures, and scenario-level statistics. These artifacts are stored locally or in a shared Drive data lake depending on the execution path.

Table IV summarizes the main datasets already available in the quantum framework work tree. The project inherits a validated quantum-routing corpus that already contains scenario-level and experiment-level outputs for four external testbeds and for the internal hybrid framework. These corpora are stored under */Users/pitergarcia/DataScience/Semester4/GA-Work/Validated_Logs* and are paired with a shared Drive data lake whose top-level folders are `model_state` and `framework_state`. This matters because the final fairness-aware paper can build directly on preserved runner/evaluator artifacts.

The DSCI601 code-review package uses a small synthetic dataset that preserves the shape of the final fairness problem without requiring private data. Each record includes a group label, signal feature, contextual feature, binary outcome, missingness flag, and invalid-record flag. The dataset is deliberately small so reviewers can run the full pipeline quickly, but it exercises the same operations needed by the final project: cleaning missing context, preparing train/test artifacts, producing predictions, computing group metrics, and writing final summaries.

The planned clinical diagnostic testbed will also be simulation-first. Its records will represent patient or case contexts, protected or fairness-relevant groups, diagnostic action choices, observed outcomes, test costs, delay or escalation signals, and injected missingness or measurement noise. This design allows controlled evaluation of false-negative and false-positive gaps before moving to any approved open or sensitive clinical data.

The clinical side also has a preliminary evidence base from the ISTE780 EQUITAS bioinformatics project [2]. That project uses RNA secondary-structure prediction as a diagnostic-relevant proxy for context-sensitive biomedical decision support. The current clinical corpus contains 20 RNA sequences after fairness-aware augmentation, including tRNA, rRNA, miRNA, synthetic controls, and 10 targeted Synthetic LowGC sequences added to correct under-representation. It records sequence type, length, GC-content bins, structural features, predicted-pair outputs, algorithm identity, and fairness-audit fields. The

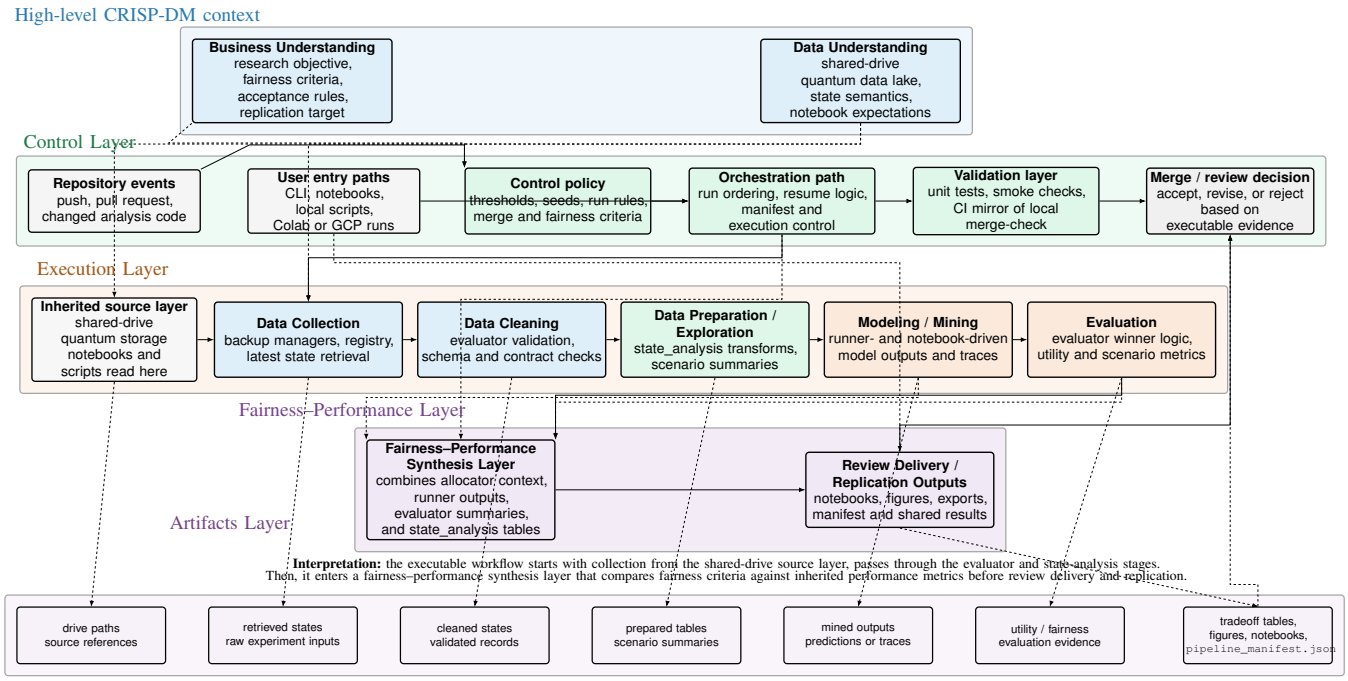

Fig. 2. Architecture diagram adapted from the architecture report. The workflow connects high-level CRISP-DM context, repository/user entry paths, control and validation layers, executable data-processing stages, the fairness–performance synthesis layer, and final replication artifacts. This diagram clarifies how notebook, local, and cloud execution paths all produce auditable evidence for review and final analysis.

TABLE IV
PRIMARY QUANTUM DATASETS AND CORPORA USED BY THE PROJECT.

Corpus	Rows	Cols	Scope	Use in this project
Hybrid master dataset	4,920	30	internal hybrid framework	fairness-layer extension on top of allocator, scenario, reward, regret, and efficiency summaries
Standardized external papers corpus	8,060	32	Papers 2, 7, 8, and 12 at 4K/2K	cross-testbed comparison baseline for adding fairness and transfer analysis
Paper 2 standardized slice	2,450	30	15 nodes, 51 edges, 8 paths	stochastic-noise external benchmark
Paper 7 standardized slice	1,710	30	50 nodes, 141 edges, 15 paths	dense-network external benchmark
Paper 8 standardized slice	1,500	30	20 nodes, 19 edges, 8 paths	compact RL-routing external benchmark
Paper 12 standardized slice	2,400	30	100 nodes, 426 edges, 4 paths	fusion-network external benchmark

DSCI601 clinical simulator will extend this evidence base into the planned sequential diagnostic workflow.

Table V summarizes the common schema. The schema is broad enough to support both testbeds but narrow enough to keep analysis comparable: each record has a decision context, a chosen action, an outcome, and a fairness-relevant group or flow label. This design supports cross-domain transfer analysis because the policy-update and mitigation logic can operate over comparable artifact contracts even when the domain-specific feature semantics differ.

The existing quantum corpora already expose many of the fields the fairness layer needs. The master datasets record allocator, scenario, scenario category, run count, frame configuration, capacity semantics, scale, model identity, reward, regret, Oracle-relative efficiency, retries, failures, misses below threshold, and scenario winners. In practical terms, this means the current paper can confidently claim access to real experimental data and preliminary results. The main extension still needed is the fairness layer itself: group or flow labels, subgroup disparity summaries, and time-evolving fairness metrics must be added on top of the already preserved performance corpus.

For the clinical side, the project remains simulation-first. The final clinical dataset will mirror the structure of the quantum corpus: context quality, action choice, binary or graded outcome, cost/delay information, and fairness- relevant group annotation. Current clinical evidence should be interpreted as preliminary and diagnostic-relevant rather than as final patient-level validation. The EQUITAS corpus demonstrates that richer

TABLE V
PLANNED DATA FIELDS AND FAIRNESS ROLE.

Field type	Purpose
Group / flow label	defines subgroup or service class for disparity metrics
Context features	patient/test signals or link/load/fidelity estimates
Missingness flags	identify degraded, delayed, or unavailable context
Action choice	selected test, escalation, model, route, or allocator decision
Outcome / reward	diagnostic success, routing success, latency, or utility
Oracle / baseline	supports regret, gap, and mitigation comparison

TABLE VI
ETHICAL DESIGN REQUIREMENTS AND IMPLEMENTATION RESPONSE.

Requirement	Implementation response
Avoid aggregate-only claims	report utility beside subgroup disparity metrics
Expose context quality	log missingness, noise, delay, and measurement assumptions
Support accountability	preserve seeds, configs, states, manifests, and run provenance
Protect sensitive data	use simulation-first clinical data until approved data are appropriate
Handle security relevance	study quantum routing under unreliable/adversarial conditions

biomedical context and fairness-aware augmentation can be measured now; the final DSCI601 clinical testbed will add the missing sequential decision layer. This keeps the cross-domain comparison meaningful. One dataset is not meant to imitate the domain semantics of the other; rather, both are designed to satisfy the same artifact contract so that the same evaluation and mitigation logic can be applied to both.

VI. ETHICS, FAIRNESS, PRIVACY, AND SECURITY

Ethics is central to the project because the system is explicitly about resource allocation under uncertainty. A policy that optimizes only aggregate reward can still be ethically unacceptable if it shifts errors onto groups that already have lower-quality context. In the clinical setting, the most direct ethical concern is harm from missed or delayed diagnosis. If context is missing because a group has less reliable testing, delayed follow-up, or lower-quality measurements, then a model can learn that the group is harder to classify and route fewer resources toward it. That is not merely a statistical artifact; it is a mechanism by which historical under-measurement becomes future under-allocation.

The project therefore treats fairness as a design requirement rather than as a post-hoc report. Fairness affects how experiments are built, how artifacts are logged, and how results are interpreted. The evaluation must report subgroup metrics, not only average reward or average efficiency. It must preserve context-quality information, because missingness is part of the causal story of disparity. It must also make assumptions visible: which groups are compared, which outcome gaps matter, which simulator mechanisms generate missingness, and which mitigation thresholds are considered acceptable. Without those details, a reproducible code artifact could still be ethically incomplete because it would hide the value choices that shape the experiment.

Table VI summarizes how ethical concerns translate into design requirements. Privacy and security are secondary analytic goals, but they still shape the project. Clinical data requires privacy-aware handling, which is why the clinical testbed is simulation-first in the current phase. Quantum routing raises a different concern: communication patterns and access to scarce network resources can be sensitive, and adversarial or unreliable conditions can change who receives successful service. The shared framework does not solve every privacy or security problem, but it creates an evaluation structure where those assumptions can be varied and documented rather than ignored.

The clinical preliminary evidence reinforces this ethical claim: low-GC and other under-represented sequence categories required explicit augmentation and re-auditing before fairness claims were defensible. The ethical standard for the final paper is therefore stronger than "the algorithm works." The final system should make visible when a policy is accurate but inequitable, robust but unfair to a service class, or fairer only because it sacrifices too much utility to be operationally useful. This tradeoff framing is important for applied data science: fairness is not a decorative metric but part of the evidence required to justify deployment or to reject a policy as unsafe.

VII. RESULTS AND CURRENT VALIDATION EVIDENCE

Full scientific results are not complete yet, but the implementation foundation has been validated. The rough-draft code package compiles and runs through the complete CRISP+ML lifecycle, and the merge-check path currently passes ten unit tests covering collection, cleaning, preparation, exploration, mining, evaluation, postprocessing, CLI behavior, and manifest generation. This is not claimed as a final scientific result; it is reproducibility evidence showing that the project has an executable review harness and that the architecture is ready for the next experimental phase.

A. Quantum Preliminary Results

The quantum side has an existing experimental foundation from prior threat-aware routing work [1]. The remaining task is to extend those performance and robustness outputs with explicit service-equity metrics and fairness-aware mitigation. Thus, current progress is strongest on infrastructure, methodology, and reproducibility, while final cross-domain fairness results remain future work.

B. Clinical Preliminary Results

The clinical testbed has preliminary evidence from the ISTE780 EQUITAS bioinformatics project [2]. This work is

TABLE VII
PRELIMINARY QUANTUM RESULTS AVAILABLE BEFORE FAIRNESS-LAYER EXTENSION.

Evidence source	Current quantitative signal	Why it matters	How it feeds this paper
Threat-aware quantum manuscript	pursuit–neural hybrids outperform non-contextual baselines by roughly 18–24 percentage points in scenario-aggregated efficiency	shows that context-rich routing materially improves utility under realistic threats	establishes the performance baseline to which the fairness layer will be added
Threat-aware quantum manuscript	worst-case performance floors remain above 85% under stochastic threats for the strongest hybrid family	suggests the best policies remain stable in benign-but-noisy settings	provides a natural starting point for testing whether fairness can be improved without breaking robust utility
Threat-aware quantum manuscript	replay-capacity increases can trigger 22–30 percentage-point efficiency collapses under adaptive adversaries	shows that resource policy, not only algorithm choice, shapes deployment quality	motivates fairness-aware evaluation of allocator and capacity decisions, not just bandit model choice
Standardized external corpus	8,060 rows across 4 testbeds, 4 allocators, 5 threat scenarios, 3 scales, and 5 routing models including Oracle	demonstrates breadth of already available quantum evidence	supports cross-testbed fairness benchmarking once subgroup/service labels are defined
Internal hybrid corpus	`iCPursuitNeuralUCB` currently has the highest mean efficiency in the internal hybrid dataset (86.13%)	identifies the strongest current utility baseline in the internal framework	provides the leading candidate for fairness-aware extension and ablation

TABLE VIII
CONTEXT-SPECTRUM SUMMARY FROM THE EXISTING GA CORPORA. GAINS ARE MEASURED AGAINST THE NO-CONTEXT FAMILY WITHIN EACH CORPUS.

Context level	Representative model(s)	Internal mean	Δ vs. no ctx.	External mean	Δ vs. no ctx.
No context	`GNeuralUCB`, `EXPNeuralUCB`	82.39%	—	62.83%	—
Context	`CPursuitNeuralUCB`	84.70%	+2.31 pp	64.11%	+1.28 pp
Predictive context	`iCPursuitNeuralUCB`	**86.13%**	**+3.74 pp**	**65.66%**	**+2.83 pp**

TABLE IX
CROSS-TESTBED PREDICTIVE-CONTEXT UPLIFT IN THE STANDARDIZED EXTERNAL CORPUS.

Testbed	No ctx.	Context	Pred. ctx.	Pred. gain
Paper 2	72.11	73.18	74.34	+2.23
Paper 7	72.87	73.44	78.39	+5.52
Paper 8	70.05	73.69	73.95	+3.90
Paper 12	41.69	42.23	42.54	+0.85

not yet the final sequential diagnostic simulator, but it supplies the clinical side of the paper with a concrete diagnostic-relevant setting: RNA structural prediction under uneven sequence/context representation. In that setting, "lower context" corresponds to simpler or less context-rich prediction signals, while "richer context" corresponds to Nussinov/MFE feature augmentation, fairness-aware synthetic low-GC augmentation, and subgroup-aware EQUITAS auditing.

Table XII frames the current evidence conservatively. It supports reproducibility and readiness, not final claims about which policy is fairest. The final paper will replace or extend this table with empirical curves, statistical summaries, and domain-specific fairness results.

Table VII makes the current evidence more concrete. The rough draft can already report real quantum results because the project builds on an existing validated experiment corpus and an associated manuscript. In the standardized 4K/2K external corpus, the four inherited testbeds contribute 8,060 rows of model–scenario–allocator–scale evaluations. In the internal hybrid corpus, the strongest current average efficiency belongs to `iCPursuitNeuralUCB` (86.13%), followed by `CPursuitNeuralUCB` (84.70%), `GNeuralUCB` (82.83%), and `EXPNeuralUCB` (81.95%). These are not yet fairness results, but they are meaningful preliminary results: they identify the current utility frontier and expose which policy families are worth extending first.

Just as importantly, the current results can already be organized according to the project's core scientific claim: *context quality matters*. Rather than treating the preliminary evidence as a list of algorithm names, Table VIII and Figure 3 recast the existing quantum framework work as evidence into a three-level spectrum. The no-context condition captures reward-only or non-context baselines, the context condition captures methods that use observed contextual structure, and the predictive-context condition captures the informative or forecasting-aware extension. In both the internal and external corpora, performance rises as the method family moves from no context to context to predictive context. That pattern is exactly what motivates our work extension: if richer context already improves utility, the next question is whether fairness-aware and missingness-aware context handling can also improve equity.

The external standardized corpus also shows that results vary by testbed structure. Mean efficiencies for `iCPursuitNeuralUCB` are currently about 74.34% on

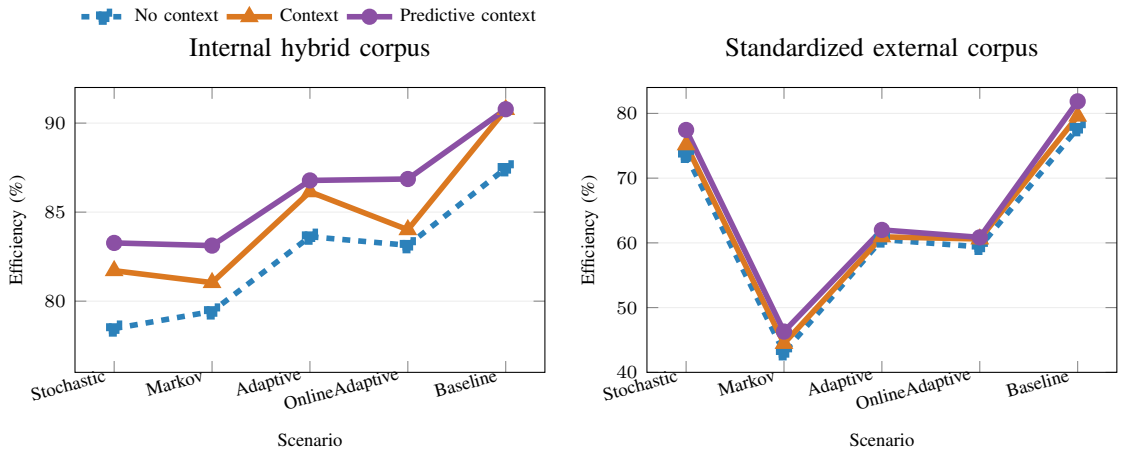

Fig. 3. Quantum-quality context-spectrum view across threat regimes. In both the internal hybrid corpus and the standardized external corpus, the predictive-context family dominates the no-context family across all five scenarios. The gain is largest in the harder external settings, where context quality more strongly determines routing efficiency.

TABLE X
PRELIMINARY CLINICAL RESULTS FROM THE EQUITAS BIOINFORMATICS TESTBED.

Evidence source	Current quantitative signal	Why it matters	How it feeds this paper
EQUITAS project corpus	dataset expanded from 10 to 20 RNA sequences, including 10 Synthetic LowGC sequences for fairness-aware augmentation	establishes that clinical-like context gaps can be made explicit and corrected in a controlled biomedical proxy task	provides the diagnostic-side dataset pattern for missingness-aware simulation
Enhanced Nussinov tuning	Optuna search over 30 trials produced an optimized objective of 56.44 and mean predicted-pair output of 66.07 $\pm$ 46.36	shows that richer structural context can materially change diagnostic-relevant prediction behavior	motivates comparing lower-context and richer-context policies in the clinical simulator
Ablation evidence	thermodynamic/context-aware scoring reported 61.36 $\pm$ 44.21 predicted pairs versus 14.96 $\pm$ 10.45 without that context	directly supports the paper's context-gain thesis in the clinical testbed	anchors the clinical side of the cross-testbed context-gain comparison
EQUITAS fairness audit	post-augmentation mitigation achieved 6/6 algorithm compliance with the 80% rule, DI ratios of 1.000, and +0.805 average DI improvement	demonstrates that representation/context repair can improve fairness indicators, not only utility	supplies preliminary fairness evidence before final sequential bandit experiments

TABLE XI
AGGREGATE CONTEXT-GAIN COMPARISON USED FOR THE CROSS-TESTBED VISUAL.

Testbed	Lower ctx.	Richer ctx.	Gain
Quantum routing	82.39%	86.13%	+3.74 pp
Clinical bioinformatics	14.96 pairs	66.07 pairs	+51.11 pairs

Paper 2, 78.39% on Paper 7, 73.95% on Paper 8, and 42.54% on Paper 12. This matters for our work because it already suggests that fairness conclusions should not be tied to a single network topology or threat setting. A fairness-aware extension that appears successful in one testbed but fails in another would still be a useful scientific finding.

From the standpoint of the rough-draft requirement, these results answer the most important preliminary question: does the system already do something real and measurable? Yes. The inherited quantum framework already executes correctly, produces large validated datasets, distinguishes strong and weak model families, and supports claims about robustness, efficiency, and configuration effects. The new contribution of the final paper is to attach an explicit fairness layer to those

TABLE XII
CURRENT VALIDATION EVIDENCE AND INTERPRETATION.

Evidence	Interpretation
CRISP+ML quick pipeline runs	reviewer can execute the code artifact end-to-end
Stage unit tests pass	lifecycle stages have isolated behavior checks
Merge-check command passes	local validation path is ready for PR gating
Quantum execution paths documented	final experiments can start from notebook, local, or GCP workflows
State architecture documented	interrupted long-running tests can be resumed or audited

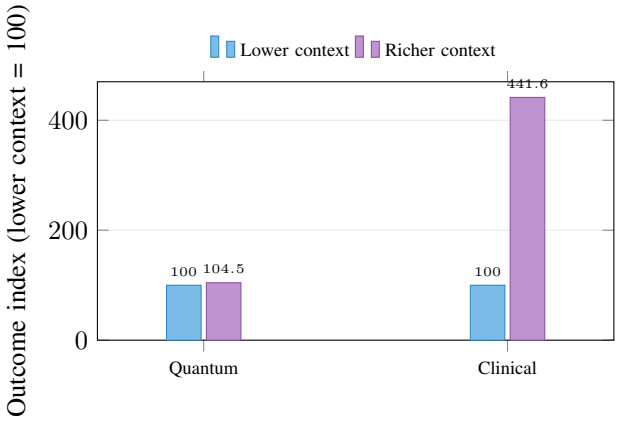

Fig. 4. Cross-testbed aggregate context-gain comparison. Values are normalized within each testbed so the lower-context condition equals 100; the graph compares the effect of adding context regardless of the specific approach used. Quantum uses Oracle-normalized efficiency, while the clinical proxy uses predicted-pair output from the EQUITAS RNA-structure evidence base.

same data and test whether the best utility policies also behave equitably under missing context.

The most important current result is that the research workflow has converged into a coherent experiment plan. Earlier pieces of the semester work existed as separate deliverables: a proposal, a related-work survey, an approach write-up, an architecture report, an ethics/fairness report, and a code-review package. This rough draft integrates those pieces into a single project narrative. The implementation now has a reviewer-safe CRISP+ML path, a quantum-first execution path, a state-management strategy, and a defined fairness layer. This reduces the risk that final experiments will produce isolated performance numbers without enough provenance to explain them.

The expected preliminary figures for the final report are also now specified. For each scenario, the analysis will report reward or efficiency curves, oracle gap, subgroup or flow-level disparity curves, and mitigation tradeoff plots. For cross-domain transfer, the analysis will compare whether a mitigation strategy that reduces missing-context disparity in one domain also reduces it in the other. If transfer fails, that failure is still informative: it would suggest that the shared framework is useful for diagnosing differences between domains even when a single mitigation rule does not generalize.

VIII. Future Work

Next, the focus will be on completing the experimental contribution. First, the quantum fairness adapter will connect evaluator and runner outputs to service-equity metrics such as success-probability and latency gaps. This will turn the existing quantum-routing performance infrastructure into a fairness-aware analysis pipeline. Second, the clinical diagnostic simulator will be implemented with controlled missingness, group-dependent measurement noise, diagnostic action choices, and error outcomes. This simulator will allow the project to test false-negative and false-positive gaps without relying on sensitive patient records during development.

Third, at least one mitigation mechanism will be added and compared against utility-only baselines. The initial priority is a simple fairness-regularized or missingness-aware update rule that can be implemented in both domains. Fourth, the same policy stack will be evaluated across both domains to test whether fairness mechanisms transfer. Fifth, the final report will add full results, plots, statistical discussion, and a stronger conclusion grounded in experimental evidence. The final analysis will also include a limitations section discussing simulation validity, group-definition choices, and the difference between fairness in a controlled testbed and fairness in deployment.

IX. Conclusion

This semester established the foundation for a reproducible fairness-aware contextual bandit project. The work completed so far includes a related-work survey, a shared missing-context domain model, an approach and architecture writeup, a cleaned code-review repository, documented notebook/local/GCP execution paths, and a runnable CRISP+ML validation package. The project is positioned to move from design and infrastructure into final experiments. Its expected impact is a practical framework for measuring and mitigating fairness disparities in sequential decision systems where context is incomplete, non-stationary, and unequally distributed across groups.

The next version of the paper will replace the current validation evidence with complete experimental results. However, the structure needed for those results is now in place: the literature gap is defined, the domain model is explicit, the architecture is reproducible, the data contracts are identified, and the ethical requirements are stated as part of the technical design. That makes the remaining work a focused execution task.

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